

Chapter I - Older Americans and Life Insurance ¹

Iris SooJin Park Tyler Q. Welch

Abstract

The conventional academic literature suggests that demand for term life insurance should decline post-retirement as income protection becomes less necessary. Yet our study reveals that 30-40% of individuals aged 65 and older maintain at least one term life policy, some even till their 80s. While employment status influences term life insurance holdings—both through income protection needs and employer-provided benefits—we find that a notable number of term life policyholders aged 65 and older hold policies despite having no life-contingent income sources, such as working income or retirement funds, to protect. This study investigates alternative rationales for holding term life insurance beyond income protection, including caregiving value, bequest motives, funeral expenses, and more. Additionally, we find that once individuals purchase term life insurance, it often becomes optimal for them to continue holding it after a short period, with this effect varying significantly by issue age. While these findings provide initial insights into older age term life insurance holdings, further analysis is needed to fully understand the relative importance of these various motivations, which we explore in this paper.

Keywords: life insurance, demand, consumption smoothing, bequests, socioeconomic status

¹Both authors based at the Wisconsin School of Business, University of Wisconsin-Madison. Park: park487@wisc.edu; Welch: tyler.welch@wisc.edu. We are grateful for helpful comments and feedback from Dani Bauer, Anita Mukherjee, Dan Sacks, Joan Schmit, and Justin Sydnor.

1 Introduction

Extensive research has explored the demand for life insurance and its determinants throughout the past century, with scholars such as Huebner (1920), Yaari (1965), Lewis (1989), Fitzgerald (1989), Li et al. (2007), and Liebenberg et al. (2012) contributing significant insights. However, these studies have rarely addressed the prevalence of term life insurance among older individuals. This omission may stem from an assumed reduction in income risk. In contrast, Brown (2001) found a notable proportion of individuals aged 70 and older maintained life insurance coverage. More than twenty years later, the continuous demand for term life insurance by the elderly remains a relatively uncharted area of study. Leveraging recent data, this study aims to answer two important questions: Do older individuals still hold term life insurance today? And if so, what motivates these older individuals to hold term life insurance?

The prevailing view in the academic field is that life insurance demand among older individuals, particularly retirees, should be minimal. As people age and transition into retirement, the income risk traditionally hedged by life insurance diminishes. Annuities and pensions replace a significant portion of an individual’s income, thereby lessening the need for term life insurance (Hambel et al., 2017). Moreover, the primary role of life insurance — to financially support dependents after the policyholder’s death — becomes less critical as the number of dependents typically declines with age (Hammond et al., 1967). Additionally, the traditional motivations for obtaining life insurance, such as mortgage coverage or new business investments, decrease with age (Huebner, 1920). Furthermore, wealth accumulation over the lifespan tends to reduce absolute risk aversion (Lin and Grace, 2007). The rising cost of premiums, driven by increased mortality and morbidity risks associated with aging, further diminishes the financial attractiveness of life insurance (Hammond et al., 1967; Hambel et al., 2017). Additionally, since term life insurance, unlike whole life insurance, does not accumulate any cash value and carries the risk of lapsing, it should theoretically further reduce its appeal as individuals age, particularly as they grow older and depend more on fixed incomes.

Despite these motivations, our research documents a contrary trend: life insurance holdings, particularly term life insurance, are quite common among older Americans, many of whom maintain any term life insurance into and beyond their 70s. We analyze responses to the Health and Retirement Survey and Survey of Consumer Finances to investigate patterns of life insurance holding at older ages. We find that 48% of Americans aged 65-69 hold term life insurance alone. For older age groups, the rates are 43% for those 70-74, 40% for 75-79, 38% for 80-84, and 32% for those 85 and older. Additionally, we observe 31% of full-time homemakers and 30% of individuals not working, receiving Social Security, and

also without any ceasing retirement income hold term life insurance. These findings challenge traditional assumptions and underscore the need for further investigation into the factors driving older individuals to maintain term life insurance coverage despite theoretical expectations of its decline; they also suggest that the demand for life insurance might not be fully explained by income protection alone, indicating other underlying motivations.

One potential reason for holding term life insurance beyond retirement age may be the structure of the policies themselves, which are typically offered in discrete terms of 5, 10, 15, or 30 years. Policyholders often cannot choose the exact year their coverage ends but must select from available fixed terms, potentially leading to coverage that extends slightly beyond retirement. This study examines whether it is beneficial for individuals to maintain their policies post-retirement and assesses the value of doing so. In order to answer this question, we estimate the actuarially fair term life insurance premium, including loading, and calculate the expected value for each term from the policyholder's perspective. Although the overall expected value of the net premium is negative for the policyholder after loading, the expected value of holding the policy each year becomes positive relatively fast. We also find a positive correlation between expected value, term length, and age. These findings may explain why there is a modest decrease in term life insurance holdings immediately after retirement and suggest that continuing the policy could be advantageous for policyholders. However, even after 5 years of retirement year, we still see approximately 45% of retired individuals have term life insurance.

To determine how much of these term life insurance holdings is explained by factors other than income protection, we limited our sample to individuals over 65, without any income that would cease upon death, including both working and annuitized income, and at least five years post-retirement if ever employed. After limiting the sample to retired or not working individuals, we found that 44% of individuals with pensions that continue still have term life insurance. This suggests that motivations likely extend beyond mere income protection. Moreover, to understand the sheer magnitude of the fraction of people that do not have any life-contingent income to protect, among those who hold any term life insurance, 47.88% have no pension other than Social Security or IRAs, and 3.78% of individuals have pensions where the amount continues after death. This means that more than half of the term life insurance holders do not have both income and life-contingent retirement funds. Therefore, in our primary analysis, we explored alternative motivations such as the value of caregiving, bequest motives, and estate tax planning. While we haven't found evidence for other motivations, our findings indicate that a significant motivation is likely the desire to leave a bequest.

In the following month, we also plan to undertake several key analyses for this paper. First, we

aim to assess whether individuals might benefit more from holding whole life insurance rather than term life insurance for bequest purposes, from the policyholder’s perspective. Second, we intend to explore additional motivations for maintaining term life insurance, such as covering burial expenses or managing debts that might be passed to a spouse or dependents. Lastly, we plan to estimate the joint effect of multiple motives on the decision to hold term life insurance, recognizing that individuals often have varied reasons for maintaining their policies. We also hope that this will help identify which motivations are most influential and whether their importance varies across different demographic groups.

In the upcoming month, we plan to undertake several key analyses to deepen our understanding of the motivations behind term life insurance holdings among older adults. First, we aim to investigate why older individuals might prefer term life insurance to simply saving money, hypothesizing that risk preferences related to bequests could drive this choice. The structured payout system of term life insurance may appeal more to those with risk-loving tendencies than the ongoing discipline required for personal savings. We plan to explore this hypothesis further by developing a utility model to explore predictions for life insurance at older ages under strong bequest motives. Second, we intend to explore additional motivations for maintaining term life insurance, such as covering burial expenses or managing debts that might be passed to a spouse or dependents. Third, after examining all potential motivations, we plan to estimate the combined effect of these various motives on the decision to hold term life insurance. To do so, we will use existing variables as proxies for motivations, such as the presence of dependents, spouse’s health status, and prior employment status. This approach will allow us to account for the complex and varied reasons individuals maintain their policies. By identifying which motivations are most influential and analyzing how their significance varies across demographic groups, we aim to provide a clearer understanding of the decision-making processes driving term life insurance holdings. Lastly, we will also briefly discuss the possibility of conducting a survey to either collect insurance information that we are currently missing from our existing dataset, such as issue age and term length, or to understand preferences related to term life insurance choices among policyholders.

In the next section, we review the literature and discuss the theoretical background in more detail. In Section 3, we discuss the HRS and SCF data and provide summary statistics. In Section 4, we document the life insurance holdings by age group; and in Section 5, we present descriptive results of term life insurance holdings and their potential underlying motivations. Lastly, in Section 6, we conclude and explain the future analysis planned.

2 Theoretical Development and Literature Review

We start with a motivation for some of the more descriptive questions we have laid out, specifically related to motivations for older individuals to purchase life insurance and the connections between work and life insurance demand. Our data allow us to provide descriptive evidence as to the theoretical predictions, including among recently retired individuals and data as current as 2018, and so we review both theoretical and empirical work in this regard. Then, in order to discuss how people utilize the life insurance policies they hold, we review work on life insurance payout utilization. For completeness, we also review the literature on how individuals use other forms of lump-sum payouts, such as those from workers' compensation or disability insurance.

2.1 Life Insurance Demand Among Older Individuals

Whether or not older individuals should demand term life insurance is an important question. Original theories for life insurance were centered around its importance in protecting the family (Huebner, 1920). This rationale implies that life insurance is based on managing the risk of losing a primary wage earner and, even with some savings, quality of life. This approach to life insurance has a necessary result: older individuals shouldn't be purchasing term life insurance (which omits the savings component of whole life insurance policies), given they are unlikely to be a primary wage earner as compared to a younger family with one working parent that is in a "saving period." As discussed above, there are other age-related reasons to expect little life insurance demand among older individuals: cost (Hammond et al., 1967), planning horizon (Levy et al., 1988), risk aversion (Lin and Grace, 2007), and - related to cost - health expenses (Hambel et al., 2017). The presence of Social Security survivors' benefits also reduces the need for term life insurance at older ages, even when there is little other annuitized wealth.

Despite the clear theoretical underpinnings for the decision, there is mixed empirical evidence. In fact, Hammond et al. (1967) observed that the relationship between age and life insurance premium expenditures varied across different income levels. In addition, it has been shown that life insurance demand is positively related to age, education, and financial vulnerability as measured by the volatility of standard living upon a spouse dying (Frees and Sun, 2010). Yet, the creators of that financial vulnerability measure ultimately found that, while the vulnerability measure was positively related to life insurance demand, the relationship dampens with age (Lin and Grace, 2007). A substantial review of age-related determination of life insurance demand notes that most signs on age are not relevant or negative, though some results are mixed and positive, providing support for the findings of Frees and Sun (2010) above

(Liebenberg et al., 2012). Even despite cohort shifts away from life insurance (Chen et al., 2001), there are many possible determinants of life insurance demand among older individuals, as nicely summarized in recent work (Cherry and Asebedo, 2022).

Using data from 1993 and 1994, Brown (2001) documented that 30 to 40 percent of individuals over age 70 held term life insurance. Their study investigated the annuity offset hypothesis, which posits that individuals who have invested heavily in annuities might purchase term life insurance to balance the risk of overannuitization. This scenario typically occurs when retirees, concerned about locking all their funds into annuities that provide stable income but cease at death, opt for term life insurance to ensure some financial benefit is passed on to their heirs, reflecting a strong bequest motive. However, contrary to the hypothesis, they found that only 0.5 percent of married men and 2 percent of widows held term life insurance potentially for this reason. Given these findings, the authors concluded that the annuity offset hypothesis does not adequately explain the still large portion of term life insurance holdings among the elderly and that alternative explanations needed further exploration. They suggested several potential motivations, including couple protection, inertia, estate tax planning, and funeral expenses. The authors noted that these alternative hypotheses had either produced mixed interpretations in existing research or had not yet been studied thoroughly. Thus, in our paper, we aim to document whether the trend of still high term life insurance holdings among the elderly, as observed by Brown, persists. Moreover, we intend to verify and test the alternative hypotheses suggested by Brown and explore some new hypotheses to see if any could adequately explain the large term life insurance holdings among the older population which the annuity offset hypothesis could not.

In the end, there seems to be little theoretical justification for older individuals, especially those not working, to purchase term life insurance (as separate from whole life insurance). This is true in part due to lifecycle factors (children are self-supporting, business development unlikely, and less likely to be working) and cost factors (term life insurance becomes substantially more expensive with age). Yet, work studying the determinants of life insurance demand do not paint as clear of a picture with respect to age and term life insurance demand. Using the HRS and SCF data at our disposal, we hope to provide descriptive evidence that renews the discussion in this literature of how prevalent term life insurance is for older individuals and provide answers to some of the possible motivations.

2.2 Connections Between Work and Life Insurance Demand

As term life insurance is primarily designed to protect against the loss of income, numerous studies have examined the relationship between life insurance demand and employment. It has been documented

that, due to the fact that employees often receive (subsidized) term life insurance coverage through an employer, there are sharp drop-offs in life insurance coverage rates along the age distribution (Miller, 1985). More recent work reviews parts of the prior literature to determine that the previously-documented empirical relationships between life insurance demand and occupation or employment indicate a positive relationship (Zietz, 2003). This is consistent with the expectation that work tends to increase life insurance demand.

The results across different types of employment also indicate important differences in demand. Specifically, self-employed individuals are more likely to demand life insurance, and spend more on term life insurance premiums, than other types of employment (Luciano et al., 2016).

Yet, what is of most interest in our setting is transitions in and out employment. There is some evidence that individuals demand term life insurance more upon new employment opportunities. Even when the evidence may not be conclusive in the reverse, there seems to be a negative relationship between new unemployment spells and term life insurance demand (Liebenberg et al., 2012). These results are not specific to older individuals, however.

When we think about the impact of retirement on life insurance demand, the first place to look is to Social Security survivors' benefits. Broadly, Social Security has plausibly little net effect on life insurance demand because survivors' benefits decrease life insurance demand (conditional on spousal death) while retirement benefits increase life insurance demand (conditional on survival of spouse), leading to offsetting effects (Fitzgerald, 1987). In the international context, there is supporting evidence that increased Social Security expenditures reduces life insurance demand (Zerriaa and Noubbigh, 2016) and that Social Security retirement benefits are positively related to private life insurance (Ma and Pope, 2008).

2.3 Life Insurance Payout Utilization

In part as we have seen above, life insurance demand is an oft-studied portion of the literature, with focuses on understanding why and when individuals demand life insurance. Our primary concern in this analysis is to better understand what individuals do with life insurance payouts, which is related to life insurance demand but also distinct from it. Naturally, the instinct we have is to investigate the use of these funds by a surviving spouse, which is useful given the surviving spouse was (presumably) involved in the decision to buy life insurance, which would include an understanding of how to spend proceeds at time of death, allowing us to study deviations from planned (or optimal) usage.²

²Though it is important to note that marital situations are not straightforward. Recent work has found that marital bargaining power is important, especially along gender lines (Wong, 2015).

Early work in this area nicely reviews bequest motives and how economists’ understanding of life insurance benefits has developed over time (Modigliani, 1988), including the development of the idea that not all wealth was developed intergenerationally but may be accumulated “from scratch by each generation.” This paper gives us, from seminal work (Keynes, 1936), that there are two important, competing motives in life insurance demand: pure bequest motives (i.e., leaving money for the next generation) and provision motives (i.e., to assist surviving spouses with consumption smoothing) – two motives that provide separate demand initiatives (Hubener et al., 2014).

In expanding the reach of Lewis (1989), Auerbach and Kotlikoff (1991) consider both the change in standards of living before and after death, and the ultimate finding that life insurance purchasing is below optimum – which finds room for Social Security policy changes. This is in line with older work finding substantial poverty impacts of widowhood (Hurd and Wise, 1987). The consumption smoothing effects we study are ex-ante ambiguous in existence and size, as noted in prior work that finds some life insurance purchases can leave widows better off than if the spouse had survived (Fitzgerald, 1989).

With this inspiration in mind, our paper follows - in part - the theory developed by Hubener et al. (2014). In that paper, the authors develop a model that pieces apart the portfolio and term life insurance choices made by a couple who may care about bequest motives or consumption smoothing for a surviving spouse. These authors’ model finds, and is empirically validated with data from the same source we use here (the HRS), that the primary reason to buy term life insurance for couples is to smooth consumption rather than to bequest. Ultimately, they identify that inequality in household income (annuitized income, too) is positively associated with term life insurance take-up, indicative of what the consumption smoothing hypothesis would suggest.

In this spirit, we aim to empirically estimate these theoretical predictions. First, we will use the HRS data to take aim at life insurance payments being used for bequests and consumption smoothing, yet we also care about other impacts of payments. Namely, we aim to assess the effect of payouts on subjective socioeconomic status, poverty benefit receipt, debt, life satisfaction, and emotional well-being. Prior work on bequests has created a debate about the relevance of bequests, where it is acknowledged that bequest motives alter the view of insurance to individuals (Bernheim, 1991), as nicely reviewed in a recent paper (Inkmann and Michaelides, 2012). Similar to that paper, we also intend to measure bequest motives, yet both before and after life insurance receipt, which has not been done in the literature (to the best of our knowledge).

2.4 Evidence from the Usage of Other Lump Sum Payouts

Outside of life insurance proceeds, there exists empirical evidence as to how individuals use lump-sum payouts from other sources, such as from retirement plans, workers' compensation insurance, and disability insurance. In general, a call for research into the usage of these lump sums has been outstanding for a significant period of time (Rea Jr., 1981). Our review of the existing literature suggests that specific evidence of consumption patterns (or bequest motives) after a lump sum payout in workers' compensation and disability insurance is lacking, and more so than in the retirement setting.³

In the retirement space, there is ample work discussing how individuals use retirement plan lump-sum payouts, which are not uncommon (Clark et al., 2014). Much of the early work from the 1980s and 1990s has been nicely summarized, including the idea that many lump-sums are spent, not saved (Poterba and Venti, 1998). Related to that, other work shows higher 401(k) balances have a lower distribution probability (Choi et al., 2004). At least from our understanding, precise documentation of consumption patterns after payout has yet to be performed in this literature, though there is some evidence – not specific to the older population – that people use their non-savings-directed payouts on direct consumption, debt, and education (Copeland, 2005).

From the workers' compensation insurance perspective, lump-sum payments are quite common (Thomason and Burton, 1993), though it is unclear whether lump-sum benefits are optimal for individuals (Schmit, 1987). There is evidence that raising workers' compensation benefits decrease consumption shocks after worker injury (Bronchetti, 2012) and that there is income heterogeneity in the response to lump sum payments, with large liquidity effects for lower earners (Rennane, 2015). Though this is not quite the same as studying how individuals use lump-sum payouts and are hard to directly relate to life insurance, the work of Bronchetti (2012) presents an important finding about consumption smoothing effects in the workers' compensation setting specifically. To our knowledge, precise evidence as to the consumption (or bequest) effects of lump sum payouts, beyond general benefit analyses on consumption smoothing, in the workers' compensation setting are sparse.⁴

In the disability insurance context, lump sum payments have also been studied. The authors of a recent paper use a regression discontinuity design to study the effect of moving from a lump sum payment to regular installments, and the results suggest that losing out on the lump sum (and receiving monthly

³There is evidence of effects of inheritances, settlements, and gifts on home ownership (Rosser, 1999), retirement (Joulfaian and Wilhelm, 1994), labor force participation that is not specific to older workers (Holtz-Eakin et al., 1993), and marginal consumption (Joulfaian, 2006).

⁴In the legal context, there has long been a concern that lump sum payments are used irresponsibly by individuals, which has been part of the push for structured settlements, but there remains little evidence of “dissipating plaintiffs” existing in reality (Babener, 2009).

installments instead) does not affect labor supply (Mullen and Rennane, 2020). Disability insurance in a more general, international context has been found to affect labor supply in a significant fashion, where increased benefits increase non-employment in Canada (Gruber, 2000), though not all academics agree as to this result (Campolieti, 2004). While not in the lump sum context specifically and based on unique data from Norway, it is established that disability insurance payments smooth consumption in a causal manner (Autor et al., 2019; Kostøl and Mogstad, 2015).

3 Data

This paper utilizes both longitudinal and cross-sectional data to examine life insurance trends among older Americans. Specifically, we draw from the Health and Retirement Study (HRS) and the Survey of Consumer Finances (SCF). The HRS, a nationally representative longitudinal survey of adults over 50 and their spouses, has been conducted biennially since 1992. It is particularly valuable for its detailed questions on life insurance and its receipt, wealth, demographics, and spousal characteristics. For this study, we focus on the period from 2002 to 2018, aligning with the introduction of several key variables. The SCF, conducted triennially by the National Opinion Research Center at the University of Chicago, provides a comprehensive cross-sectional view of financial and life insurance data. Unlike the HRS, the SCF does not specifically target older adults, yet its detailed financial variables and its definite distinction between term life insurance and whole life insurance policies are instrumental for our analysis. For consistency, we use SCF data spanning from 2001 to 2019.

A key distinction between the HRS and SCF datasets in this study lies in how term life insurance is defined. The SCF directly inquires whether respondents have term life insurance and requests details on its face value. HRS is valuable in its detail on questions such as beneficiaries, information on spouse and dependents, ability to link respondent’s and spouses information, and its panel structure. However, the HRS only asks about the total number of life insurance policies and the number of whole life insurance. Consequently, we define a term life insurance indicator in the HRS as any instance where the total number of life insurance policies exceeds the number of whole life policies. This approach assumes that any additional policies are term life insurance. Despite this assumption, we demonstrate in Section 4 that the proportion of term life insurance holdings is similar across the two datasets. While the SCF provides a clearer definition of term life insurance, we leverage both datasets to capitalize on the longitudinal structure of the HRS, which allows us to examine changes in term life insurance holdings in relation to shifts in certain life events.

In Table 1 below, we present summary statistics for the individuals aged 50 and over in the HRS and SCF. Here, we display information before and after individuals turn 65. Given that the HRS is a panel dataset, this summary statistics table is emblematic of our panel dataset where people are permitted to age across the columns over our sample period (2002-2018). In that way, the number of observations listed is really the number of person-wave observations.

Table 1: Summary Statistics Merged: HRS and SCF Data

	HRS		SCF	
	50-64	65-85+	50-64	65-85+
Income (\$1000s)	114.07 (1.180)	71.56 (0.963)	156.57 (2.550)	96.05 (2.691)
Median Income (\$1000s)	41.47	73.21	47.47	80.26
Male Fraction	0.48 (0.002)	0.44 (0.002)	0.74 (0.002)	0.64 (0.002)
Married Fraction	0.72 (0.002)	0.60 (0.002)	0.60 (0.002)	0.50 (0.002)
Working for Pay Fraction	0.63 (0.002)	0.16 (0.001)	0.69 (0.002)	0.16 (0.002)
Hours Worked	40.02 (0.067)	29.40 (0.131)	31.32 (0.089)	6.87 (0.073)
Age	58.04 (0.014)	74.75 (0.025)	56.60 (0.017)	74.72 (0.035)
Race - White Fraction	0.79 (0.002)	0.87 (0.001)	0.74 (0.002)	0.82 (0.002)
Race - Black Fraction	0.11 (0.001)	0.09 (0.001)	0.14 (0.001)	0.11 (0.002)
Race - Hispanic Fraction	0.10 (0.001)	0.07 (0.001)	0.08 (0.001)	0.04 (0.001)
Race - Other Fraction	0.09 (0.001)	0.04 (0.001)	0.04 (0.001)	0.02 (0.001)

Note: Depicted are sample averages (or medians, in the case of the second and fourth rows) across variables (rows) and within age bands (columns). Standard errors are in parentheses. Respondent survey weights used. Wealth and income variables measured in 2018 U.S. dollars. Source: HRS 2002-2018 and SCF 2001-2019.

Our analysis yields several key insights from the income data presented in the table. First, while both the HRS and SCF datasets show similar income figures, the SCF data is consistently slightly higher across age brackets. Despite a general decline in average income as individuals move from the 50-64 age bracket to the 65-85+ bracket, median income increases for those aged 65 and over. This rise is largely due to a shift from variable employment earnings to more stable, fixed incomes from pensions and Social Security. Furthermore, individuals in lower-income brackets often retire as soon as they become eligible for benefits, typically out of necessity. This early transition to fixed incomes helps stabilize median income levels among the older age group, leading to a more uniform distribution of income.

Second, in both the HRS and SCF datasets, racial distribution remains relatively similar across age groups, though the SCF shows a slightly higher fraction of males compared to HRS. Racial and ethnicity-related mortality gaps are also evident, with the proportion of White individuals increasing in the older age group as the representation of Black and Hispanic individuals declines. This trend aligns with previous findings on racial life expectancy gaps (Dwyer-Lindgren et al., 2022). Although some research projects that Hispanic individuals may have the longest life expectancy, a detailed decomposition by race and ethnicity is outside the scope of this study.

In terms of gender and marital status, the data shows a noticeable decrease in the proportion of males and married individuals after age 65. This decline can largely be attributed to women’s higher life expectancy, which results in a greater number of females in older age brackets, and the increase in widowhood as mortality rates rise. These shifts in gender and marital status have important economic implications, as they affect household income, benefit eligibility, and the need for support services among the elderly.

4 Older Americans and Life Insurance Holdings

In this section, we examine the distribution of term life insurance holdings for Americans over the age of 50. Our analysis confirms the general understanding noted in previous research that life insurance holdings decrease with age. However, we also find that despite this negative relationship, the level of term life insurance remains notably high among older age groups.

In Table 2, it becomes clear that holding life insurance in old age is not at all uncommon. While the prevalence of purchasing new coverage is decreasing with age and represents only 6% at its peak (after Social Security retirement age), there is still a large swath of people for whom term life insurance is especially persistent into old age. In the 70-74 age bucket, when individuals must at least have a substantial piece of annuitized income from Social Security (and any Individual Retirement Accounts or IRAs, due to the required minimum distributions), 50% of individuals still hold life insurance. 33% of those aged 70-74 hold only term life insurance, with 7% having only whole life insurance and 10% with a combination of the two. To us as analysts, this is surprising for several reasons. Perhaps the most glaring is that life insurance’s pure insurance value is diminished at extremely old ages, yet 38% of individuals aged 85 and above hold life insurance when just 2% of our sample works for pay at those ages. Even between ages 70-74, when 17% of our sample works for pay, life insurance holdings are substantial (and 4% of the sample is buying new coverage at these ages). Among individuals aged 80-84, 30% hold only

term life insurance. That percentage remains elevated at 26% for those aged 85 and older.

Table 2: Life Insurance Holdings by Age Group

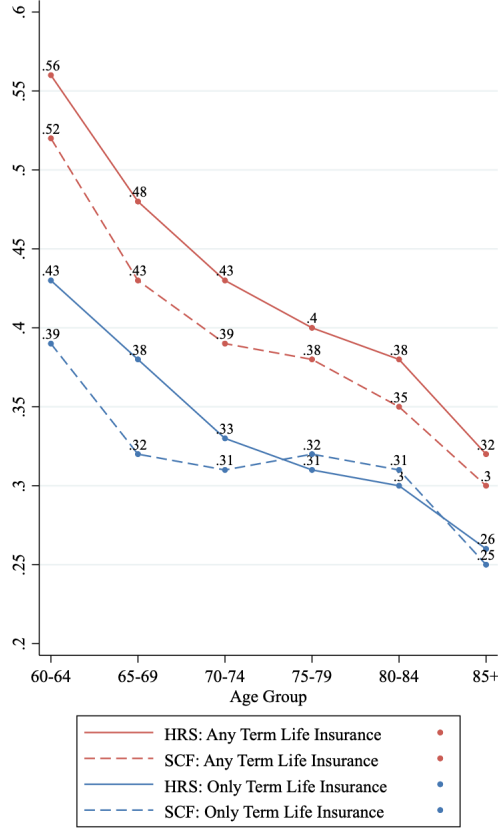
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	50-54	55-59	60-64	65-69	70-74	75-79	80-84	85+
Any Life Insurance	0.62 (0.005)	0.65 (0.003)	0.61 (0.003)	0.54 (0.004)	0.50 (0.004)	0.48 (0.004)	0.47 (0.005)	0.38 (0.005)
Any Term Life Insurance	0.59 (0.005)	0.61 (0.003)	0.56 (0.004)	0.48 (0.004)	0.43 (0.004)	0.40 (0.004)	0.38 (0.005)	0.32 (0.005)
Only Term Life Insurance	0.48 (0.005)	0.48 (0.003)	0.43 (0.004)	0.38 (0.004)	0.33 (0.004)	0.31 (0.004)	0.30 (0.005)	0.26 (0.005)
Both Term and Whole Life Insurance	0.11 (0.003)	0.13 (0.002)	0.12 (0.002)	0.10 (0.002)	0.10 (0.002)	0.09 (0.002)	0.08 (0.003)	0.06 (0.002)
Only whole Life Insurance	0.03 (0.002)	0.04 (0.001)	0.05 (0.002)	0.06 (0.002)	0.07 (0.002)	0.09 (0.002)	0.09 (0.003)	0.07 (0.003)
Obtained new life insurance	0.11 (0.004)	0.08 (0.002)	0.07 (0.002)	0.06 (0.002)	0.04 (0.002)	0.03 (0.002)	0.02 (0.002)	0.01 (0.001)
<i>N</i>	9,601	21,183	19,980	18,582	17,022	14,466	10,279	8,891

Notes: Depicted are sample averages across variables (rows) and within age bands (columns). Standard errors are in parentheses. Respondent survey weights used. Source: HRS 2002-2018.

While life insurance holdings shown in Table 2 are based on HRS dataset, it is also important to note that the term life insurance holdings reported in HRS duplicates the results from SCF significantly. In Figure 1, we can see that the percentage of each age group that holds “any or only term life insurance” nearly duplicates across data sources. Detailed distribution with SCF dataset is also shown in Table A.1 as well. Thus, from here, we can conclude that the term life insurance holdings among the older population is observed in both datasets and that is likely that term life insurance fraction from HRS is not an overestimation of the holdings.

Additionally, to understand why older Americans purchase term life insurance from an industry perspective, we also surveyed 48 financial advisors. Of these, 91.67% sell life insurance or are certified to do so, and 70.83% have had clients aged 70 and above. Among the advisors who have older clients, they reported that on average, 21.28% of these clients hold term life insurance. This statistic, derived from 25 responses, is not a comprehensive representation of term life insurance holdings among the elderly. Additionally, this fraction might be smaller than what is observed in the general elderly population as reported in Table 2, since it pertains only to individuals who have financial advisors. Nevertheless, it serves as an indication that term life insurance among the elderly is a notable trend observed by professionals in the field.

Figure 1: Term Life Insurance Holdings by age group: HRS and SCF



Notes: This figure compares term life insurance holdings by age group from two datasets: HRS (solid line) and SCF (dotted line). Red lines indicate any term life insurance, and blue lines indicate only term life insurance. Source: HRS 2002-2018, SCF 2001-2019

5 Results and Motivations of Term Life Insurance Holdings

In this section, we analyze the descriptive patterns and characteristics of term life insurance holdings to understand the underlying motives of policyholders. By examining various demographic and financial factors, we aim to identify the reasons that drive individuals to opt for term life insurance, providing insights into their decision-making processes.

First, we look at respondent-year observations across our panel (as in Table 1), Table 3 displays summary statistics across categories of life insurance policyholders. We group policyholders into a few mutually exclusive categories: column (1) includes individuals who do not have life insurance in a wave and column (2) includes those individuals who do have life insurance (of any type) in a wave. Among those respondent-wave observations included in column (2), we further study those individual-wave observations with only whole life insurance in column (4), and those with both term and whole life insurance in column (5). Throughout much of our paper, we will discuss those with only term life insurance, which is depicted

in column (3).

Table 3: Life Insurance Holdings

	(1) None	(2) Any Type	(3) Only Term	(4) Only Whole	(5) Both Term & Whole
Household Wealth (\$1000s)	573.54 (6.453)	622.92 (5.826)	530.78 (5.812)	958.07 (24.579)	796.29 (16.861)
Median Household Wealth (\$1000s)	164.73	260.20	224.40	377.69	359.65
Household Income (\$1000s)	70.18 (1.901)	111.28 (0.669)	104.44 (0.732)	116.13 (2.543)	134.31 (1.731)
Median Household Income (\$1000s)	36.07	73.13	70.50	63.03	87.56
Years of Education	12.39 (0.015)	13.53 (0.011)	13.39 (0.013)	13.52 (0.031)	14.06 (0.023)
Male	0.40 (0.002)	0.52 (0.002)	0.49 (0.002)	0.58 (0.006)	0.58 (0.005)
Married	0.56 (0.002)	0.71 (0.002)	0.69 (0.002)	0.72 (0.005)	0.77 (0.004)
Working for Pay	0.27 (0.002)	0.53 (0.002)	0.54 (0.002)	0.35 (0.006)	0.58 (0.005)
Hours Worked	34.10 (0.138)	39.61 (0.080)	39.66 (0.093)	36.19 (0.335)	40.56 (0.177)
Age	67.60 (0.045)	64.51 (0.038)	64.00 (0.045)	69.00 (0.119)	64.12 (0.083)
White	0.82 (0.002)	0.85 (0.001)	0.85 (0.002)	0.86 (0.004)	0.88 (0.003)
Black	0.09 (0.001)	0.10 (0.001)	0.10 (0.001)	0.11 (0.004)	0.09 (0.003)
Other Race	0.09 (0.001)	0.05 (0.001)	0.05 (0.001)	0.03 (0.002)	0.04 (0.002)
Hispanic	0.14 (0.001)	0.05 (0.001)	0.06 (0.001)	0.03 (0.002)	0.03 (0.002)
<i>N</i>	56,628	63,376	44,401	7,124	11,851

Notes: Depicted are sample averages (or medians, in the case of the second and fourth rows) across variables (rows) and within life insurance holding categories (columns). Standard errors are in parentheses. Respondent survey weights used. Wealth and income variables measured in 2018 U.S. dollars. Source: HRS 2002-2018.

As is clear in this table, life insurance holdings broadly are associated with more wealth and household income (both in terms of means and medians). Those with life insurance tend to also be more educated, slightly younger, workers with longer work weeks, male, married, working for pay, and White or Black (relative to the other race category). Those with life insurance are also much less likely to be Hispanic.

Among those who have life insurance (columns (3) through (5), for reference), those with only whole life insurance policies are substantially wealthier than those with only term life insurance and those with both policies. The story for household income is less clear, especially looking at the median, likely highlighting that term life insurance holders are much more likely to be working for pay (and to be working more on the intensive margin), where they gain access to term life insurance as an employee benefit. The distributions of education, marriage, race, and ethnicity are similar. Those with only term life insurance do tend to be much more likely to be female and younger.

There is an important fact that reveals itself in this table. As discussed above in the motivation of this paper, there is lesser insurance value coming from term life insurance as people age and stop working, given the protection of one’s non-annuitized income becomes less important as individuals retire and receive annuities (from pensions and/or Social Security) or tap into defined contribution retirement assets. Of the people in our sample holding term life insurance (any or only), only just over half are working at the time they hold it, less than 80% are married, and the average age indicates people are nearing Social Security eligibility. This indicates that some in our sample are holding term life insurance at times that they do not have working income and less concern about financial needs.

5.1 Life Insurance, Employment, and Annuitized Income

In order to understand how employment status impacts term life insurance holdings, this section reports on life insurance data by various employment statuses and examines if there is a strong correlation between term life insurance coverage and the exact timing of retirement. Table 4 displays life insurance holdings information across respondent-years for individuals who are employed, unemployed (and searching for work), retired, and full-time homemakers⁵. The data indicate that employed individuals are significantly more likely to have term life insurance or life insurance in general compared to those who are unemployed or retired. While only 24% of unemployed individuals and 43% of retirees hold term life insurance, nearly 63% of employed individuals have at least one term policy. Some of this difference could be explained by two factors: primarily, term life insurance may have been initially purchased for income protection; secondarily, the high fraction may reflect the employer-provided term life benefits that provide coverage during their employment. However, we also need to focus on the fact that the level of term life insurance holdings of retired individuals or homemakers is relatively large. From here, we can infer that while income protection does serve as a major motivation for the general population, it does not explain the majority of the term life insurance holdings after their retirement age.

5.2 5-Year Term Structure and Value of Keeping

One possible explanation for why a significant portion of retirees continue to hold term life insurance even after their retirement year could be the structured increments in which these policies are issued. Term life insurance is typically available in 5-year increments. Therefore, someone enrolling at age 49 and seeking coverage until at least age 65 might choose a 20-year term that would inadvertently extend coverage to age 69. This structuring could result in retirees maintaining coverage beyond their initial

⁵Individuals who reported being homemaker for the entire sample period

Table 4: Life Insurance Status relative to Employment and Retirement Status

	Employed	Unemployed	Retired	Year of Retirement	Homemaker
Any Life Insurance	0.68 (0.002)	0.27 (0.008)	0.51 (0.002)	0.54 (0.008)	0.35 (0.008)
Any Term Life Insurance	0.63 (0.002)	0.24 (0.008)	0.43 (0.002)	0.47 (0.008)	0.31 (0.008)
Only Term Life Insurance	0.49 (0.002)	0.21 (0.008)	0.34 (0.002)	0.36 (0.008)	0.27 (0.008)
Only Whole Life Insurance	0.05 (0.001)	0.03 (0.003)	0.07 (0.001)	0.07 (0.004)	0.05 (0.004)
<i>N</i>	42,312	2,771	54,980	3,764	3,292

Notes: Depicted are sample averages across variables (rows) by employment and retirement status. Standard errors are in parentheses. Respondent survey weights used.

retirement age, addressing financial protection concerns that extend into later life stages.

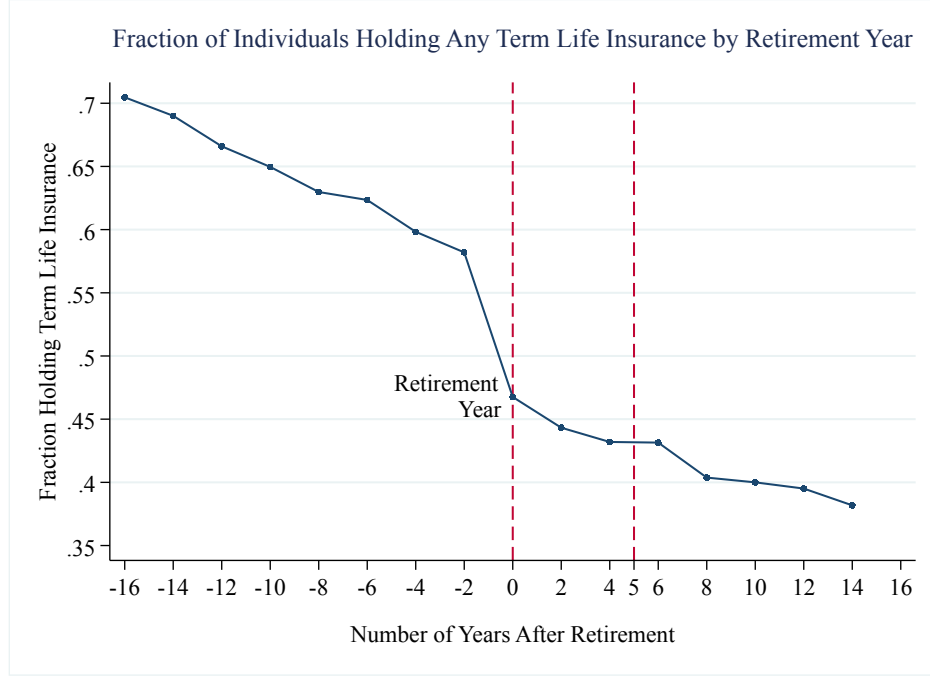
Unfortunately, datasets like the HRS or SCF do not provide specific details on the start or end ages of term life insurance policies. However, the longitudinal structure of HRS allows us to investigate whether there is a noticeable decline in term life insurance holdings within five years following retirement. If the standard 5-year increments in term life insurance influenced individuals' decisions regarding their insurance holdings and if individuals had a certainty about their retirement age, we would expect to see a reduction in the fraction of term life insurance five years post-retirement.

In Figure 2, we present the proportion of term life insurance holdings by the number of years post-retirement among respondents who reported retiring during the survey period. We observe a significant decline in term life insurance holdings in the year of retirement. Further, the figure also indicates a decrease of 8% drop within the first 7 years and 5% drop within the first 5 years. While we anticipated a more significant reduction between the fourth and sixth years, the slight delay observed also could be attributed to inaccuracies in predicting the precise retirement age at policy enrollment.

This consideration leads to a crucial follow-up question: "If individuals are keeping term life insurance after retirement because of the discrete structure, would individuals benefit from canceling the policy after retiring?" Exploring this question requires an understanding of how premiums are calculated and the mortality risk dynamics as the policyholder ages. Premiums for term life insurance are calculated based on the mortality rate averaged over the entire term of the policy. This approach is designed to spread the risk of mortality evenly across the term, which makes the overall expected cost of the insurance initially appear negative for the policyholder after considering the loadings for the insurer.

However, as a policyholder ages and the remaining term of the policy shortens, the mortality rate applicable to the remaining years often exceeds the initial average. For example, a policyholder who initially purchased a 5-year term life policy at age 45 would pay for a premium that is calculated with

Figure 2: Fraction of Individuals Holding Any Term Life Insurance by Retirement Year



Note: This figure plots the fraction of any term life insurance holdings by number of years after retirement. The sample used for this plot was restricted to individuals who reported retiring during the survey period. Source: HRS 2002-2018

an average mortality rate between ages 45 and 50. However, after 2 years, the applicable mortality rate for the policy years between ages 47 and 50 is higher than the initial average mortality rate used for calculating the premium. This increase in mortality risk can make continuing term life insurance more beneficial for policyholders as the policy matures, contrary to the decision to cancel it post-retirement.

To determine when the expected value of term life insurance becomes positive, and to explore how it varies by issue age, term length, and face value, we begin by estimating the actuarially fair premium for each set of risk levels. The calculations are based on the equations described in *Life Insurance Mathematics, Third Edition*, by Gerber (2013).

Survival Probability

$$P(\text{Death at age } x) = q_x = \frac{\text{Mortality per 1000}}{1000} \quad (1)$$

$$P(\text{Survive at age } x) = p_x = 1 - q_x \quad (2)$$

$${}_k p_x = p_x \times p_{x+1} \times p_{x+2} \times \cdots \times p_{x+k-1} \quad (3)$$

where:

- k is the year of the term life insurance.
- x is the age at enrollment.

Net Premium

$$\text{Net premium} = \frac{\text{EPV}_{\text{death benefits}}}{\text{EPV}_{\text{premiums}}} = \frac{\sum_{k=0}^{n-1} v^{k+1} \cdot {}_k p_x \cdot q_{x+k} \cdot \text{face value}}{\sum_{k=0}^{n-1} v^k \cdot {}_k p_x} \quad (4)$$

where:

- x is the age at enrollment.
- k is the year of the term life insurance.
- i is the interest rate.
- v is the discount factor, which is defined as $v = \frac{1}{1+i}$

Using established equations, we calculate term life insurance premiums for individuals aged 40 to 75, with coverage amounts ranging from \$10,000 to \$300,000, increasing in \$50,000 increments. We consider terms ranging from 5 to 30 years, with increments of 5 years. Combinations where the term ends outside of ages 65 and 95 were excluded, resulting in a total of 1,057 combinations. We assumed an arbitrary interest rate of 5%. For the calculation of survival probabilities, we employ the 2017 Commissioners Standard Ordinary (CSO) Tables from the Society of Actuaries, specifically the 2017 Loaded CSO Composite Gender-Blended Tables. These tables provide age-specific mortality rates for each policy term. The inclusion of a loading factor serves two critical purposes: it compensates for possible variances in mortality experiences among different insurers and ensures reserve adequacy by adding a margin to mortality rates.

Below in Table 5, we present a summary of monthly premiums for term life insurance, categorized by each term length and issue age, with a face value of \$200,000 (Premiums for other face values were also calculated, but not reported for simplicity). As expected, the estimated premium shows a negative correlation with both issue age and term length.

Table 5: Average Monthly Premiums by Issue Age and Term Duration for a Face Value of \$200,000

Issue Age	term Y1	term Y5	term Y10	term Y15	term Y20	term Y25	term Y30
40-44	5.18	10.78	16.79	24.01	33.12	44.40	55.08
45-49	8.19	16.50	26.36	38.56	53.66	71.99	88.89
50-54	10.89	27.32	44.14	63.77	87.84	117.46	143.71
55-59	15.84	43.01	71.70	103.23	143.20	189.71	228.03
60-64	26.67	65.27	105.93	157.10	222.92	295.10	345.81
65-69	37.81	95.28	159.04	241.10	349.37	445.53	428.92
70-74	54.86	138.58	251.21	403.98	557.14	558.87	
75-79	71.11	176.19	336.17	548.00	710.48		

Notes: Premiums for each face value was calculated for this study, but for simplicity, we only report the monthly premium for a face value of \$200,000.

Having estimated the premiums, we now calculate the overall expected value of the policies and

the expected value for each term year. The calculation of the forward-looking expected value assesses the policy's value each year, under the assumption that the policyholder has survived to that age after making the required premium payments. This method calculates the conditional expected value at each age, beginning from the current age or any chosen point in the policy's timeline. We estimate future expected values by considering two main scenarios: the possibility of paying out a death benefit if the policyholder passes away within the year, and the continuation of the policy into the following year if they survive.

Overall Expected Value

First, the overall expected value can be calculated as following; and by the definition of net premium, the overall expected cost should be zero with the loaded mortality used to calculate the premium. Additionally, if we calculate the overall expected value with pure mortality, the overall expected value for the policyholder would be negative.

$$\text{Overall Expected Value} = \underbrace{\sum_{k=0}^{n-1} {}_k p_x \cdot q_k \cdot FV \cdot \left(\frac{1}{1+i}\right)^k}_{\text{Expected PV of Death Benefit}} - \underbrace{\sum_{k=0}^{n-1} {}_k p_x \cdot \text{premium} \cdot \left(\frac{1}{1+i}\right)^{k-1}}_{\text{Expected PV of Premium Collected}} = 0 \quad (5)$$

Forward Looking Expected Value

Unlike the overall expected value calculation, the expected value calculated here is conditioned on the individual keeping the policy up to that point. As a result, the expected value might turn positive in these cases because, although the premium is initially based on the average mortality rate for the entire period, the average mortality rate for the remaining period could be higher if the policyholder has already been holding the policy for some time. This reflects the increasing risk of death as the policyholder ages, which wasn't fully captured at the initial premium setting.

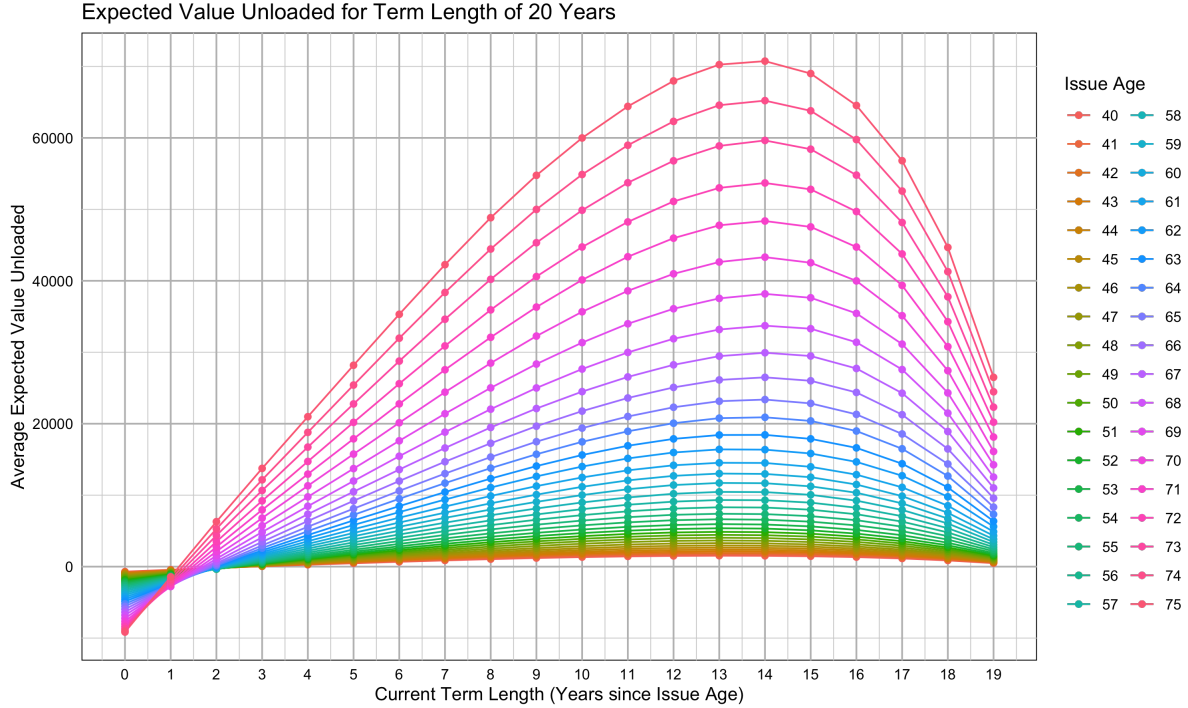
$$V(a) = \begin{cases} q_a \cdot FV \cdot \frac{1}{1+i} - \text{premium}, & \text{if } a = \text{max age} \\ q_a \cdot FV \cdot \frac{1}{1+i} - \text{premium} + (1 - q_a) \cdot \frac{1}{1+i} \cdot V(a+1), & \text{if } a < \text{max age} \end{cases} \quad (6)$$

where a is current age and q_a is the mortality at age a .

Below in Figure 3 and A.1, we present the estimated expected values by term year and issue age for each term length with a face value of \$200,000. Initially, the expected value for the first year or two is negative for the policyholder, indicating an initial cost from loading factors. However, this value quickly turns positive as the years progress, suggesting that if the policyholder retains the policy beyond these

early years, it becomes financially beneficial for them to continue holding the policy. Additionally, there is a notable increase in expected value as the issue age increases. This positive correlation between issue age and expected value suggests that the benefits of maintaining term life insurance are larger if they issue the policy at older ages, primarily due to the increasing mortality rate associated with aging.

Figure 3: Expected Value by Term Year and Issue Age for 20Y Term



Note: This figure displays the forward looking expected value by each term year and by issue age. The premiums were calculated for a 20-year term life insurance policy with a face value of \$200,000. An interest rate of 5% was used to calculate the premiums.

From the analysis above, we observe that the expected value of term life insurance often becomes positive within a few years of policy issuance, underscoring that maintaining the policy can be financially advantageous beyond initial years. This shift suggests that while the primary motivation for purchasing term life insurance may initially be income protection, the benefits of continuing the policy extend into later life stages. However, even taking into consideration that there could be a lag in term life insurance holdings after retirement age, a large part of life insurance holdings cannot be explained by income protection alone. This is particularly evident among groups like full-time homemakers and individuals retired for a longer period who maintain coverage despite having no active employment income to protect. In the next section, I further investigate other potential reasons that could motivate people to buy term life insurance that extends into their older ages, exploring factors beyond mere income replacement.

5.3 Annuitized Income

Another possible income source that individuals might be covering with term life insurance is their annuitized income. For individuals without retirement savings or who rely on annuitized incomes that would cease upon their death, term life insurance could serve to smooth income for the surviving spouse.

Multiple sources of annuitized income, such as Social Security, IRAs, 401(k)s, and pensions, play a crucial role in the retirement strategies of individuals aged 65 or older. To assess whether the demand for term life insurance could be driven by the need to protect retirement funds, it is important to understand how these funds are typically handled upon the policyholder’s death—whether they continue to benefit the spouse or reduces. Benefits from sources like Social Security typically transfer to the surviving spouse upon death, providing continuous financial support and reducing the need for term life insurance as a retirement income smoothing tool. Similarly, while not all older individuals possess IRAs or 401(k)s, these accounts generally allow spousal access post-death. Thus, if the primary purpose of term life insurance was to protect retirement funds, we would expect low holdings among this group.

To better understand term life insurance holdings among those without working income, we focus our dataset exclusively on individuals aged 65 or older who are retired, homemakers, or disabled. Additionally, we only include those who have been retired for at least five years (see section 5.2). Table 6 illustrates the life insurance holdings for respondents who receive Social Security, IRA/Keogh plans, or have no additional pension. To ensure that the sample in Table 6 focuses on individuals without any additional funds that disappear with death, we exclude respondents with pensions (or veteran benefits) that cease or reduce after death.

Contrary to expectations, 30% and 38% of individuals receiving retirement funds from Social Security or IRA/Keogh plans respectively hold term life insurance. Additionally, 31% of those without any additional pension also hold term life insurance. This suggests that approximately 30% of individuals without any working or retirement income to protect still opted for term life insurance.

Then would individuals who have a retirement fund that would disappear with death more likely to hold term life insurance? Unlike Social Security or IRA/Keogh, pension benefits can either stop, reduce, or continue after the individual’s death. Similarly, Veteran Benefits are halved and given to spouse upon the veteran’s death, which could justify holding term life insurance for income protection, aligning with the retirement income smoothing objective.

If term life insurance were primarily used to smooth retirement income, we would expect higher coverage rates among individuals whose pensions decrease or cease after their death. Conversely, lower

Table 6: Life Insurance Status relative to Retirement Fund

	Receive Social Sec	IRA or Keogh	No Pension
Any Life Insurance	0.37 (0.007)	0.47 (0.005)	0.37 (0.003)
Any Term Life Insurance	0.30 (0.007)	0.38 (0.005)	0.31 (0.003)
Only Term Life Insurance	0.25 (0.006)	0.28 (0.004)	0.26 (0.003)
Only whole Life Insurance	0.06 (0.004)	0.09 (0.003)	0.06 (0.001)
<i>N</i>	4,647	10,239	27,421

Note: Sample for this table is restricted for individuals who are 65 or older, and retired for more than 5 years, disabled, or homemaker. Respondents who have a pension (or veteran benefit) that either stop or reduce were excluded from the sample. Depicted are sample averages across variables (rows) by Pension Receipts. Standard errors are in parentheses. Respondent survey weights used. Source: HRS 2002-2018.

Table 7: Life Insurance Status relative to Pension and Veteran Benefits

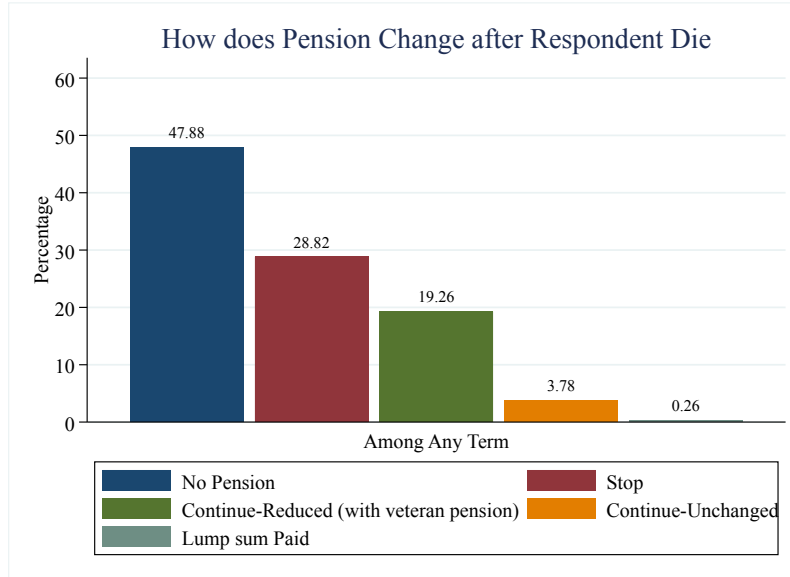
	Pension				Veteran Benefits	
	No Pension	Stop	Unchange	Reduced	Self	Spouse
Any Life Insurance	0.38 (0.003)	0.55 (0.005)	0.54 (0.012)	0.64 (0.007)	0.55 (0.008)	0.57 (0.015)
Any Term Life Insurance	0.32 (0.003)	0.48 (0.005)	0.44 (0.012)	0.54 (0.008)	0.45 (0.008)	0.47 (0.015)
Only Term Life Insurance	0.26 (0.003)	0.38 (0.004)	0.34 (0.012)	0.40 (0.008)	0.33 (0.008)	0.36 (0.014)
Only Whole Life Insurance	0.06 (0.001)	0.07 (0.002)	0.10 (0.007)	0.10 (0.005)	0.10 (0.005)	0.10 (0.009)
<i>N</i>	28,007	12,124	1,605	4,110	3,686	1,103

Note: Sample for this table is restricted to individuals who are 65 or older, retired for more than 5 years, disabled, or homemakers. The table shows sample averages across variables (rows) by Pension and Veteran Benefits. Standard errors are in parentheses. Respondent survey weights used. Source: HRS 2002-2018.

rates would be anticipated for those with unchanged pension benefits. Table 7 reveals that nearly half of the individuals facing a reduction or cessation of pension benefits after their death maintain term life insurance policies. Additionally, the likelihood of holding term life insurance is 10% higher among individuals with reducing pension benefits compared to those whose benefits remain unchanged. However, the Table also shows that 44% of individuals whose pension benefits do not change also maintain term life insurance, with 34% holding only term policies.

In the case of the Veteran’s Benefits, the need for retirement income smoothing varies depending on whether the respondent or their spouse is the veteran. As mentioned above, Veterans’ benefits are halved after the veteran’s death. Therefore, term life insurance holdings are justifiable for respondents who are veterans themselves, as their death directly impacts the household’s financial stability by reducing these benefits. While individuals receive the same benefit when the spouse is the veteran, their own death does not result in a loss of retirement income, making the first case—with the respondent as the veteran—with a much higher demand for term life insurance. Despite this, Table 7 shows a larger proportion of respondents with veteran spouses holding term life insurance compared to those whose death would directly reduce their partner’s retirement income.

Figure 4: Distribution of Pension Characteristics among Term Life Insurance Holders



Note: This figure is based on a sample restricted to individuals who are 65 or older, retired for more than 5 years, disabled, or homemakers. It illustrates the distribution of pension status among those holding any term life insurance. Individuals with veteran benefits are included in the Continue-Reduced group. Source: HRS 2002-2018.

These findings raise important questions about the motivations behind purchasing term life insurance among individuals with stable retirement funds and no employment income. Are these policyholders even

a significant segment of the market? Figure 4 illustrates the distribution of pension characteristics among term life insurance holders. From the perspective of consumption smoothing, it is notable that 47.88% of individuals without any pension and 3.78% of individuals whose pensions will continue after death hold term life insurance. These statistics suggest that nearly half of the policyholders cannot be explained by the need to smooth income or retirement income and that they might have other motives that we do not yet understand. The lack of sensitivity in term life insurance demand to changes in employment or pension status among older individuals suggests an intriguing scenario: term life insurance is being purchased even when there is no apparent risk of income discontinuity, such as from work or retirement funds. This implies that other, less understood motives for holding term life insurance may exist among this demographic, beyond the widely accepted consumption smoothing rationale.

5.4 Value of Care

Now that we have observed from the previous section that a noticeable fraction of term life insurance holdings cannot be explained by employment status or pensions alone, we turn our attention to other potential motivations. While income typically ceases upon death, other valuable contributions such as care and time also cease, which can have significant implications for surviving family members. For instance, if an individual was providing care for their spouse, their death could create a significant care gap. In such scenarios, term life insurance could serve a crucial role by funding nursing home care or a caregiver, effectively protecting against the loss of the insured's productive time devoted to care.

Table 8 presents the fraction of life insurance holdings under two scenarios: cases where a married couple reported being the primary caregiver for their partner, and cases where one partner has difficulties with Activities of Daily Living (ADLs) such as bathing, dressing, mobility, using the toilet, and eating, while the other does not. Our hypothesis is that if term life insurance is used to safeguard the productive use of time and care for a spouse, then we would expect to see a higher demand for insurance among spouses who act as caregivers. To minimize the confounding effect of retirement income smoothing, we further restrict our sample to individuals who do not receive a pension that would diminish or cease upon death.

In the first two columns, we can see that any or only term life insurance holdings are exactly the same between the caregivers and receivers. This result could indicate that the value of care might not be influencing the older population to hold term life insurance. What if we look at cases when one partner is experiencing difficulty with daily tasks while the other is not? If we compare column 3 and 4, we can see that it is a little more likely for the respondent to hold life insurance if they are sick compared to the

case when their spouse is sick. This is not only contrary to the above hypothesis but also is very minimal to point out. The pattern of term holding is also same even if we focus on the cases when the difference between the couple is larger.

Table 8: Life Insurance Holdings by Caregiving Role within Couples

	Helper Role		ALD 0	ADL ≥ 1		ADL ≥ 3	
	Care Giver	Care Receiver		SP sick	R sick	SP sick	R sick
Fraction of any life insurance	0.46 (0.010)	0.44 (0.009)	0.48 (0.005)	0.45 (0.013)	0.47 (0.012)	0.41 (0.021)	0.41 (0.019)
Fraction of any term life insurance	0.36 (0.010)	0.36 (0.009)	0.40 (0.005)	0.35 (0.012)	0.38 (0.012)	0.33 (0.020)	0.35 (0.018)
Fraction of only term	0.29 (0.009)	0.29 (0.008)	0.31 (0.004)	0.28 (0.012)	0.30 (0.011)	0.26 (0.019)	0.26 (0.017)
Fraction of only whole	0.09 (0.006)	0.08 (0.005)	0.08 (0.002)	0.09 (0.007)	0.09 (0.007)	0.08 (0.011)	0.07 (0.009)
N	2,422	2,908	11,773	1,484	1,701	560	686

Note: This table is based on a sample restricted to individuals who are 65 or older, retired for more than five years (if ever employed), disabled, or homemakers. Additionally, the sample excludes individuals without life contingent retirement income. Standard errors are in parentheses. Respondent survey weights used. Source: HRS 2002-2018.

However, when examining health status as a determinant of term life insurance demand, it's critical to focus on health conditions at the time of purchase rather than current health status. While a spouse's illness - or their own - might appear to motivate insurance holdings, this explanation becomes less compelling if the policy was purchased well before any health issues arose. For instance, if term life insurance was acquired 20 years ago, but spousal health problems only emerged in the past 10 years, this timing mismatch could indicate that the caregiving protection hypothesis was not the initial reason why individual purchased the term life insurance at th first place. Moreover, as demonstrated in Section 5.1, individuals generally benefit from maintaining their term life insurance regardless of subsequent health changes for themselves or their spouse. To properly evaluate whether caregiving value drives term life insurance purchases, we need to analyze spousal health conditions at the time of policy initiation. Thus, I plan to further explore this relationship and also layout some initial plans on doing so in the final section.

5.5 Bequest and estate tax

Bequest Motive

Another common hypothesis of term life insurance demand is bequest motive. While term life insurance do have the risk of lapsing before death, individuals still might consider term life insurance as a way of leaving money for their children.

Table 9 reports life insurance holdings by marital status and the presence of dependents. The *De-*

pendent column identifies individuals who financially support a family member for more than half of that member’s needs. The data demonstrate that having children or dependents correlates with higher levels of term life insurance holdings: married individuals show a 12% higher level, divorced individuals a 5% higher level, and widowed individuals an 8% higher level compared to those without children. For those supporting a highly dependent family member, the disparity in term life insurance holdings is even more pronounced, with married showing a 19% higher level, divorced 10% higher, and widowed 16% higher.

These findings suggest significant associations, yet they do not confirm that the bequest motive is the sole reason for purchasing term life insurance. However, individuals may have initially considered the dual purpose of income smoothing and providing a bequest from the outset. This foresight might have led them to purchase term life insurance policies that provide coverage extending well beyond their retirement age well into their 70s and 80s.

Table 9: Life Insurance Holdings by Marital Status and Dependents

	Married			Divorced			Widowed		
	No Child	Child	Dependent	No Child	Child	Dependent	No Child	Child	Dependent
Fraction of any life insurance	0.41 (0.028)	0.55 (0.004)	0.59 (0.015)	0.35 (0.038)	0.41 (0.009)	0.47 (0.036)	0.37 (0.029)	0.45 (0.005)	0.52 (0.023)
Fraction of any term life insurance	0.34 (0.027)	0.46 (0.004)	0.53 (0.015)	0.33 (0.038)	0.38 (0.009)	0.43 (0.035)	0.31 (0.028)	0.39 (0.005)	0.47 (0.023)
Fraction of only term	0.27 (0.025)	0.34 (0.003)	0.44 (0.015)	0.27 (0.036)	0.32 (0.008)	0.39 (0.035)	0.28 (0.027)	0.31 (0.005)	0.37 (0.022)
Fraction of only whole	0.07 (0.014)	0.09 (0.002)	0.06 (0.007)	0.02 (0.011)	0.04 (0.003)	0.04 (0.014)	0.06 (0.015)	0.07 (0.003)	0.05 (0.010)
Obtained new life insurance	0.02 (0.013)	0.03 (0.002)	0.06 (0.010)	0.00 (0.009)	0.04 (0.005)	0.03 (0.017)	0.04 (0.020)	0.03 (0.003)	0.06 (0.014)
<i>N</i>	307	19,606	1,101	155	3,126	198	281	9,382	475

Note: This table is based on a sample restricted to individuals who are 65 or older, retired for more than five years (if ever employed), disabled, or homemakers. Additionally, the sample excludes individuals without life contingent retirement income. Standard errors are in parentheses. Respondent survey weights used. Source: HRS 2002-2018.

In order to distinguish between the intent to leave a bequest and the initial protection against loss of working income, I focus on individuals who have been homemakers for the entire sample period. Table 10 presents the life insurance holdings among this specific group; importantly, these individuals did not have the motive of protecting working income at the time of enrollment. Among these homemakers, we still observe approximately 30% holding term life insurance. Furthermore, the data indicate that the likelihood of holding term life insurance slightly increases with the number of children they have.

As I progress in my research, I intend to concentrate on the bequest motive for term life insurance ownership, especially examining its significance among the older demographic. I also hypothesize that individuals may exhibit risk-loving behaviors when considering leaving a bequest. To delve deeper into this, I’m considering developing a utility function to explore why someone might choose to purchase term life insurance rather than saving the equivalent amount of the premium in a bank. While these thoughts

Table 10: Life Insurance Holdings - Among Fulltime Homemakers

	Homemaker	
	With Kid	With More Than 3 Kids
Fraction of any life insurance	0.37 (0.011)	0.39 (0.013)
Fraction of any term life insurance	0.31 (0.010)	0.33 (0.013)
Fraction of only term	0.27 (0.010)	0.29 (0.012)
Fraction of only whole	0.06 (0.005)	0.06 (0.006)
<i>N</i>	2,019	1,322

Note: Individuals were counted as homemakers if they reported being homemakers for the entire sample period. The sample also excludes individuals without life contingent retirement income. Standard errors are shown in parentheses. Respondent survey weights used. Source: HRS 2002-2018.

are still in the preliminary stages, I aim to develop a more detailed plan shortly.

Estate Tax

Another important aspect to consider is the tax treatment of life insurance proceeds. Generally, these proceeds are not subject to income tax, which strategically positions life insurance as a beneficial tool for managing estate tax liabilities, especially when estates consist of illiquid assets. This is particularly relevant for business owners, whose assets are often not readily convertible to cash. A survey conducted in 1997 highlighted that more than two-thirds of family business owners anticipated life insurance to serve as the primary source of funds to cover estate taxes ([Anderson, 1997](#)).

The purchase of term life insurance could be beneficial if

$$W_{\text{death with LI}} > W_{\text{death wo LI}} \quad (7)$$

where

$$W_{\text{death with LI}} = B + L + 1(\text{lapse_flag} = 0) \cdot FV - \text{Total Premium} - T - w(T - (L + FV \cdot 1(\text{lapse_flag} = 0))) \quad (8)$$

and

$$W_{\text{death wo LI}} = B + L - T - w(T - L) \quad (9)$$

for

- B is Business Asset
- L is Liquid Asset

⁶this equation is based on the equations provided by [Holtz-Eakin et al. \(2001\)](#)

- T is Total Estate Tax
- FV is the Face Value of Term Life Insurance Policy
- w is the multiplier representing the additional cost or penalty of liquidating non-liquid asset/business

While further refinement and adjustment of the equations may be necessary, we could assess the optimality of term life insurance holdings using financial variables from the SCF dataset to estimate the wealth returns both with and without life insurance policies. Descriptively, we can also analyze the changes in term life insurance holdings among high-asset individuals around the significant 2018 increase in estate tax exemption from \$5.49 million to \$11.18 million. If term life insurance were primarily purchased for estate tax mitigation, we would expect to see a reduction in these holdings among high-asset households post-2018.

Table 11: Estate Tax Exemption Over the Years

Year	Estate Tax Exemption (Single)	Estate Tax Exemption (Married)
2014	\$5,340,000	\$10,680,000
2015	\$5,430,000	\$10,860,000
2016	\$5,450,000	\$10,900,000
2017	\$5,490,000	\$10,980,000
2018	\$11,180,000	\$22,360,000
2019	\$11,400,000	\$22,800,000
2020	\$11,580,000	\$23,160,000
2021	\$11,700,000	\$23,400,000
2022	\$12,060,000	\$24,120,000
2023	\$12,920,000	\$25,840,000
2024	\$13,610,000	\$27,220,000

Source: Estate tax, Internal Revenue Service (IRS).

However, it is crucial to consider that the current estate tax exemption limits are considerably high. By 2024, the exemption limits have reached \$13.61 million for singles and \$27.22 million for married couples. Such high thresholds are unlikely to be the primary driver behind the substantial holdings of term life insurance, which range from 20% to 30%. In fact, in our sample between 2002 to 2018, in the HRS dataset, only 2.36% of individuals who hold any term had asset that was above estate tax exemption limit. Additionally, [Holtz-Eakin et al. \(2001\)](#) also found that while business owners are more likely to purchase insurance, there is no substantial evidence that they use life insurance extensively to offset estate tax liabilities. This suggests that factors other than estate tax planning likely influence the prevalence of term life insurance among these individuals.

6 Conclusion and Future Analysis

The analysis conducted thus far, as detailed in Section 4, reveals that a significant number of older individuals, in their 60s, 70s, and even 80s, continue to hold term life insurance. In Sections 5.1 and 5.3, we have demonstrated that while employment significantly influences term life insurance holdings, it does not completely account for the sustained high demand among those in their late 70s and 80s. Notably, we found that over 50% of term life insurance holders do not have any income or retirement funds that cease upon death. Additionally, using forward-looking expected value calculations, we demonstrated in Section 5.2 that after the initial one or two years of purchasing term life insurance, it is financially advantageous for individuals to maintain their policies, with the expected value varying significantly by the age at which the policy was issued.

Thus, in Sections 5.4 and 5.5, we explored non-financial motivations for purchasing term life insurance. Although we found no definitive evidence regarding the value of care, there is a strong indication that having children or dependents significantly motivates people to purchase more term life insurance. However, further analysis of these reasons, along with an exploration of other potential motivations not yet considered, is essential. In the concluding section, I outline additional analyses that we plan to conduct and propose some that could be considered to better understand the reasonings behind this term life insurance purchase.

Appendices

A Additional Tables and Figures

Table A.1: Life Insurance Holdings by Age Group - SCF Data

	(1) 50-54	(2) 55-59	(3) 60-64	(4) 65-69	(5) 70-74	(6) 75-79	(7) 80-84	(8) 85+
No Life Insurance	0.30 (0.003)	0.31 (0.003)	0.30 (0.003)	0.36 (0.004)	0.37 (0.005)	0.35 (0.005)	0.38 (0.007)	0.46 (0.008)
Any Life Insurance	0.70 (0.003)	0.69 (0.003)	0.70 (0.003)	0.64 (0.004)	0.63 (0.005)	0.65 (0.005)	0.62 (0.007)	0.54 (0.008)
Any Term Life Insurance	0.60 (0.003)	0.56 (0.003)	0.52 (0.004)	0.43 (0.004)	0.39 (0.005)	0.38 (0.005)	0.35 (0.007)	0.30 (0.007)
Only Term Life Insurance	0.47 (0.003)	0.43 (0.003)	0.39 (0.004)	0.32 (0.004)	0.31 (0.004)	0.32 (0.005)	0.31 (0.006)	0.25 (0.007)
Only Whole Life Insurance	0.10 (0.002)	0.13 (0.002)	0.18 (0.003)	0.21 (0.003)	0.24 (0.004)	0.27 (0.005)	0.27 (0.006)	0.24 (0.007)
<i>N</i>	21,639	21,103	18,674	14,505	10,839	7,884	5,193	3,820

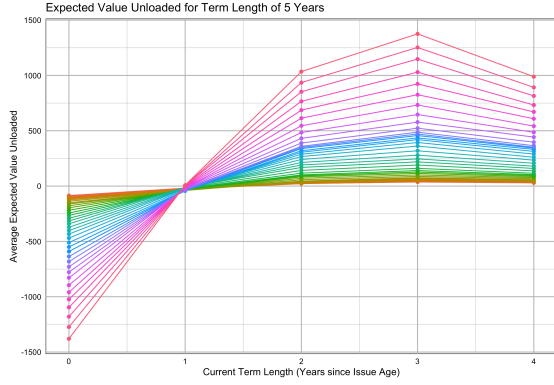
Notes: Depicted are sample averages across variables (rows) and within age bands (columns). Standard deviations are in parentheses. Respondent survey weights used. Source: SCF 2001-2019.

Table A.2: Beneficiaries of any term by marital status and dependents

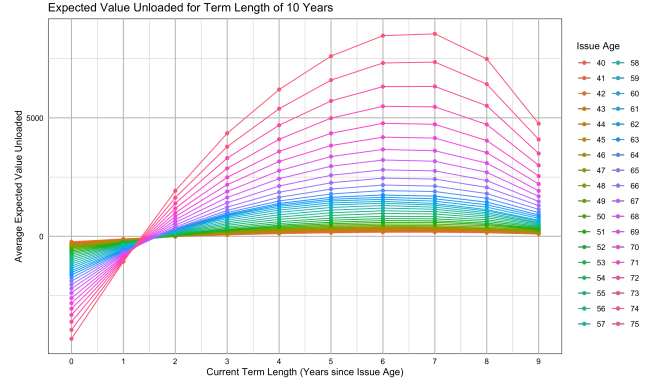
	Married			Divorced			Widowed		
	No Child	Child	Dependent	No Child	Child	Dependent	No Child	Child	Dependent
R'S SPOUSE/PARTNER	0.95 (0.021)	0.90 (0.003)	0.92 (0.011)	0.00 (0.000)	0.01 (0.002)	0.00 (0.007)	0.01 (0.011)	0.00 (0.001)	0.02 (0.008)
All Children	0.00 (0.000)	0.11 (0.003)	0.10 (0.013)	0.01 (0.014)	0.40 (0.014)	0.35 (0.049)	0.00 (0.000)	0.38 (0.008)	0.28 (0.030)
All Grandchildren	0.02 (0.015)	0.01 (0.001)	0.01 (0.004)	0.02 (0.019)	0.04 (0.005)	0.03 (0.018)	0.02 (0.015)	0.02 (0.002)	0.05 (0.014)
R's Siblings	0.06 (0.023)	0.00 (0.001)	0.00 (0.003)	0.43 (0.072)	0.01 (0.003)	0.03 (0.018)	0.15 (0.040)	0.01 (0.001)	0.00 (0.000)
Other	0.07 (0.025)	0.02 (0.001)	0.02 (0.006)	0.55 (0.073)	0.08 (0.008)	0.25 (0.045)	0.70 (0.051)	0.08 (0.004)	0.10 (0.020)
<i>N</i>	107	8,885	552	48	1,250	94	81	3,733	224

Note: This table presents the designated beneficiaries for term life insurance holders. If a holder's spouse or partner is listed as a beneficiary on seven life insurance policies considered, 'R'S SPOUSE/PARTNER' is recorded as 1. A similar notation applies to other beneficiaries. Standard deviations are in parentheses. Respondent survey weights used. Source: HRS 2002-2018.

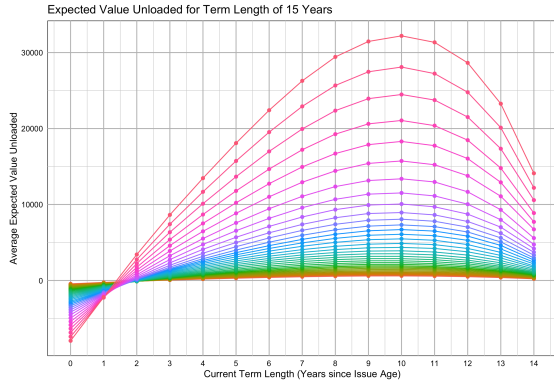
Figure A.1: Expected Value by Term Year for Each Term Length



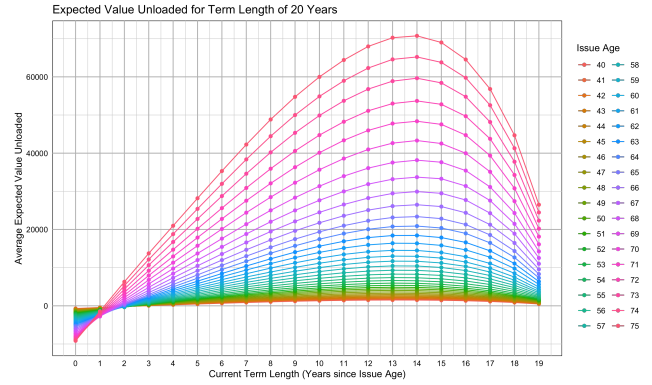
(a) 5 Year Term



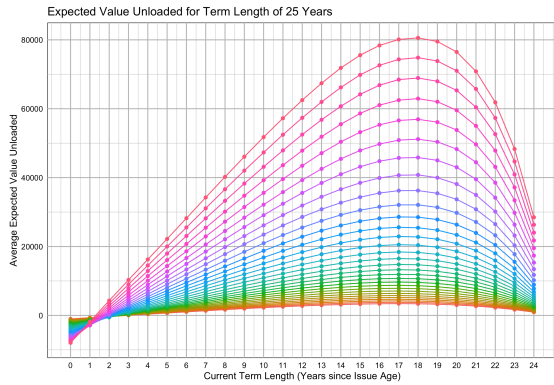
(b) 10 Year Term



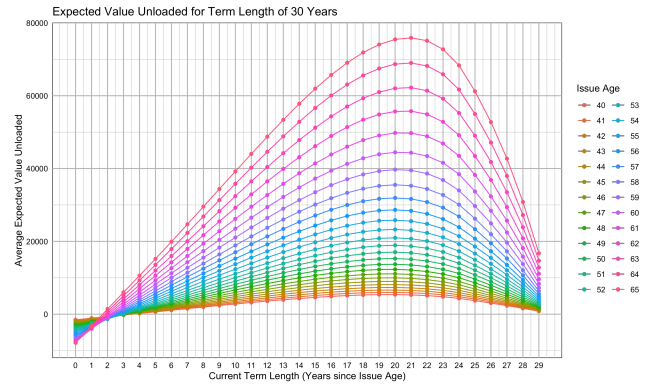
(c) 15 Year Term



(d) 20 Year Term



(e) 25 Year Term



(f) 30 Year Term

Note: This figure displays the forward looking expected value by each term year and by issue age. The premiums were calculated for a 20-year term life insurance policy with a face value of \$200,000. An interest rate of 5% was used to calculate the premiums.

References

- Anderson, A. (1997). The arthur andersen/mass mutual american family business survey= 97. *Arthur Andersen Center for Family Business and Mass Mutual: The Blue Chip Company*.
- Auerbach, A. J. and L. J. Kotlikoff (1991). The Adequacy of Life Insurance Purchases. *Journal of Financial Intermediation* 1(3), 215–241.
- Autor, D., A. Kostøl, M. Mogstad, and B. Setzler (2019). Disability Benefits, Consumption Insurance, and Household Labor Supply. *American Economic Review* 109(7), 2613–2654.
- Babener, J. (2009). Justifying the Structured Settlement Tax Subsidy: The Use of Lump Sum Settlement Monies. *New York University Journal of Law Business* 6(1), 127–168.
- Bernheim, B. D. (1991). How Strong Are Bequest Motives? Evidence Based on Estimates of the Demand for Life Insurance and Annuities. *Journal of Political Economy* 99(5), 899–927.
- Bronchetti, E. T. (2012). Workers’ compensation and consumption smoothing. *Journal of Public Economics* 96, 495–508.
- Brown, J. R. (2001). Are the Elderly Really Over-Annuitized? New Evidence on Life Insurance and Bequests. In D. A. Wise (Ed.), *Themes in the Economics of Aging*, pp. 91–124. Chicago, IL: University of Chicago Press.
- Campolieti, M. (2004). Disability Insurance Benefits and Labor Supply: Some Additional Evidence. *Journal of Labor Economics* 22(4), 863–889.
- Chen, R., K. A. Wong, and H. C. Lee (2001). Age, Period, and Cohort Effects on Life Insurance Purchases in the U.S. *Journal of Risk and Insurance* 68(2), 303.
- Cherry, P. D. and S. Asebedo (2022). Personality Traits and Life Insurance Ownership Among Older Americans. *Journal of Personal Finance* 21(2).
- Choi, J. J., D. Laibson, and B. C. Madrian (2004). Plan Design and 401(k) Savings Outcomes. *National Tax Journal* 57(2), 275–298.
- Clark, R. L., M. S. Morrill, and D. Vanderweide (2014). Defined benefit pension plan distribution decisions by public sector employees. *Journal of Public Economics* 116, 73–88.
- Copeland, C. (2005). Lump-Sum Distributions. *EBRI Notes* 26(12), 7–17.
- Dwyer-Lindgren, L., P. Kendrick, Y. O. Kelly, D. O. Sylte, C. Schmidt, B. F. Blacker, F. Daoud, A. A. Abdi, M. Baumann, F. Mouhanna, E. Kahn, S. I. Hay, G. A. Mensah, A. M. Nápoles, E. J. Pérez-Stable, M. Shiels, N. Freedman, E. Arias, S. A. George, D. M. Murray, J. W. Phillips, M. L. Spittel, C. J. Murray, and A. H. Mokdad (2022). Life expectancy by county, race, and ethnicity in the USA, 2000–19: a systematic analysis of health disparities. *The Lancet* 400, 25–38.
- Fitzgerald, J. M. (1987). The Effects of Social Security on Life Insurance Demand by Married Couples. *Journal of Risk and Insurance* 54(1), 86–99.
- Fitzgerald, J. M. (1989). The Taste for Bequests and Well-Being of Widows: A Model of Life Insurance Demand by Married Couples. *Review of Economics and Statistics* 71(2), 206–214.
- Frees, E. W. and Y. Sun (2010). Household Life Insurance Demand: A Multivariate Two-Part Model. *North American Actuarial Journal* 14(3), 338–354.

- Gerber, H. U. (2013). *Life insurance mathematics*. Springer Science & Business Media.
- Gruber, J. (2000). Disability Insurance Benefits and Labor Supply. *Journal of Political Economy* 108(6), 1162–1183.
- Hambel, C., H. Kraft, L. S. Schendel, and M. Steffensen (2017). Life Insurance Demand Under Health Shock Risk. *Journal of Risk and Insurance* 84(4), 1171–1202.
- Hammond, J. D., D. B. Houston, and E. R. Melander (1967). Determinants of Household Life Insurance Premium Expenditures: An Empirical Investigation. *Journal of Risk and Insurance* 34(3), 397–408.
- Holtz-Eakin, D., D. Joulfaian, and H. S. Rosen (1993). The Carnegie Conjecture: Some Empirical Evidence. *Quarterly Journal of Economics* 108(2), 413–435.
- Holtz-Eakin, D., J. W. Phillips, and H. S. Rosen (2001). Estate taxes, life insurance, and small business. *Review of Economics and Statistics* 83(1), 52–63.
- Hubener, A., R. Maurer, and R. Rogalla (2014). Optimal Portfolio Choice with Annuities and Life Insurance for Retired Couples. *Review of Finance* 18(1), 147–188.
- Huebner, S. S. (1920). Life Insurance in Its Relation to Thrift. *The ANNALS of the American Academy of Political and Social Science* 87(1), 183–189.
- Hurd, M. D. and D. Wise (1987). The Wealth and Poverty of Widows: Assets Before and After the Husband’s Death. *NBER Working Paper*.
- Inkmann, J. and A. Michaelides (2012). Can the Life Insurance Market Provide Evidence for a Bequest Motive? *Journal of Risk and Insurance* 79(3), 671–695.
- Joulfaian, D. (2006). Inheritance and Saving. *NBER Working Paper*.
- Joulfaian, D. and M. O. Wilhelm (1994). Inheritance and Labor Supply. *Journal of Human Resources* 29(4), 1205–1234.
- Keynes, J. (1936). *The General Theory of Employment, Interest, and Money*. New York, NY: Harcourt, Brace & Company.
- Kostøl, A. R. and M. Mogstad (2015). Earnings, Disposable Income, and Consumption of Allowed and Rejected Disability Insurance Applicants. *American Economic Review: Papers & Proceedings* 105(5), 137–141.
- Levy, H., J. L. Simon, and N. A. Doherty (1988). Optimal term life insurance - A practical solution. *Insurance: Mathematics and Economics* 7(2), 81–93.
- Lewis, F. D. (1989). Dependents and the Demand for Life Insurance. *American Economic Review* 79(3), 452–467.
- Li, D., F. Moshirian, P. Nguyen, and T. Wee (2007). The Demand for Life Insurance in OECD Countries. *Journal of Risk and Insurance* 74(3), 637–652.
- Liebenberg, A. P., J. M. Carson, and R. E. Dumm (2012). A Dynamic Analysis of the Demand for Life Insurance. *Journal of Risk and Insurance* 79(3), 619–644.
- Lin, Y. and M. F. Grace (2007). Household Life Cycle Protection: Life Insurance Holdings, Financial Vulnerability, and Portfolio Implications. *Journal of Risk and Insurance* 74(1), 141–173.

- Luciano, E., J. F. Outreville, and M. Rossi (2016). Life Insurance Ownership by Italian Households: A Gender-Based Differences Analysis. *Geneva Papers on Risk and Insurance - Issues and Practice* 41, 468–490.
- Ma, Y.-L. and N. Pope (2008). Foreign Share, Insurance Density, and Penetration: An Analysis of the International Life Insurance Market. *Risk Management and Insurance Review* 11(2), 327–347.
- Miller, M. A. (1985). Age-related reductions in workers' life insurance. *Monthly Labor Review* 108(9), 29–34.
- Modigliani, F. (1988). The Role of Intergenerational Transfers and Life Cycle Saving in the Accumulation of Wealth. *Journal of Economic Perspectives* 2(2), 15–40.
- Mullen, K. J. and S. Rennane (2020). Understanding the Impact of Cash on Hand on the Labor Supply of Disabled Workers. *NBER Working Paper*.
- Poterba, J. M. and S. F. Venti (1998). Lump-Sum Distributions from Retirement Saving Plans: Receipt and Utilization. In D. A. Wise (Ed.), *Inquiries in the Economics of Aging*, The Economics of Aging, pp. 85–108. Chicago, IL: University of Chicago Press.
- Rea Jr., S. A. (1981). Lump-Sum versus Periodic Damage Awards. *Journal of Legal Studies* 10(1), 131–154.
- Rennane, S. (2015). Buying Time: The Insurance Value and Distortionary Effects of Workers' Compensation. *Working Paper*.
- Rosser, M. (1999). Determinants of graduates' demand for housing. *Applied Economics Letters* 6(3), 139–142.
- Schmit, J. T. (1987). Lump-Sum Awards in Workers' Compensation. *Journal of Risk and Insurance* 54(2), 332–340.
- Thomason, T. and J. F. Burton, Jr. (1993). Economic Effects of Workers' Compensation in the United States: Private Insurance and the Administration of Compensation Claims. *Journal of Labor Economics* 11(1), S1–S37.
- Wong, E. S. (2015). Marital bargaining in the demand for life insurance: evidence from the Health and Retirement Study. *Review of Economics of the Household* 13, 243–268.
- Yaari, M. E. (1965). Uncertain Lifetime, Life Insurance, and the Theory of the Consumer. *Review of Economic Studies* 32(2), 137–150.
- Zerriaa, M. and H. Noubbigh (2016). Determinants of Life Insurance Demand in the MENA Region. *Geneva Papers on Risk and Insurance - Issues and Practice* 41, 491–511.
- Zietz, E. N. (2003). An Examination of the Demand for Life Insurance. *Risk Management and Insurance Review* 6(2), 159–191.